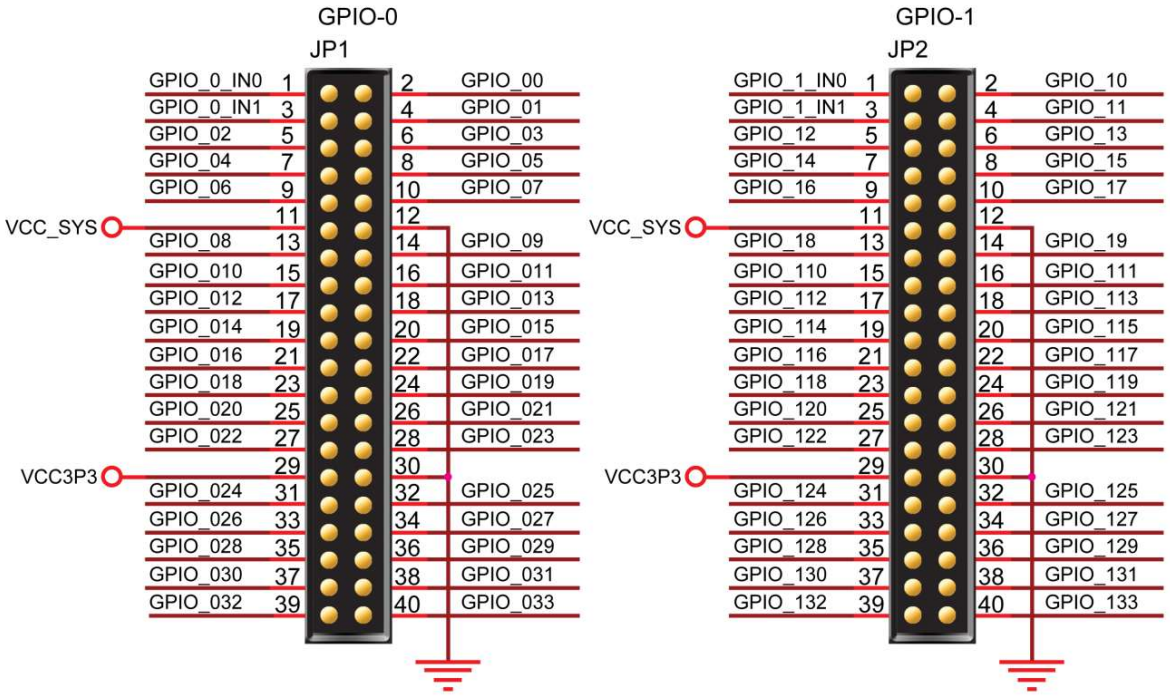


Figure 3-8 Pin arrangement of the GPIO expansion headers

The pictures below indicate the pin 1 location of the expansion headers.

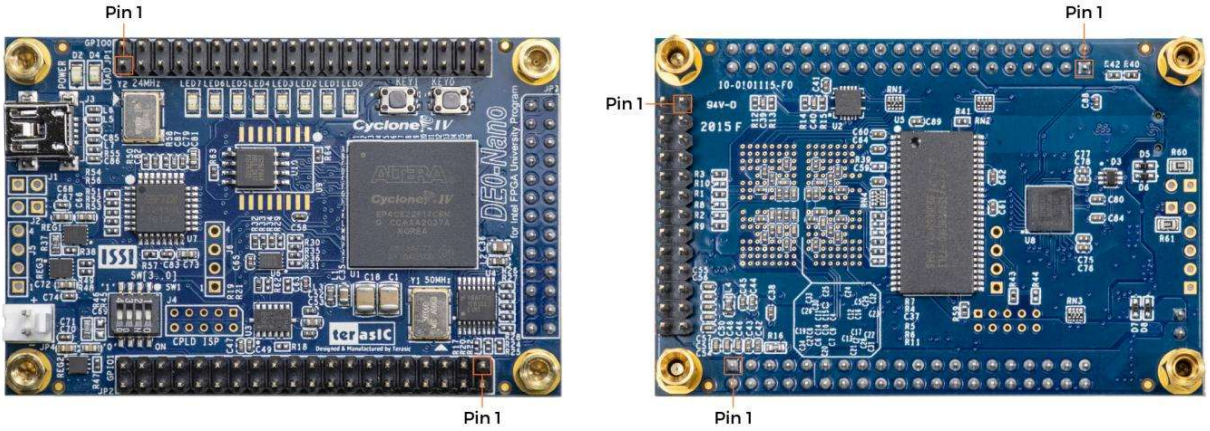


Figure 3-9 Pin1 locations of the GPIO expansion headers

Table 3-6 GPIO-0 Pin Assignments

Signal Name	*FPGA Pin No.*	*Description*	*I/O Standard*
GPIO_0_IN0	PIN_A8	GPIO Connection DATA	3.3V
GPIO_00	PIN_D3	GPIO Connection DATA	3.3V
GPIO_0_IN1	PIN_B8	GPIO Connection DATA	3.3V
GPIO_01	PIN_C3	GPIO Connection DATA	3.3V

GPIO_02	PIN_A2	GPIO Connection DATA	3.3V
GPIO_03	PIN_A3	GPIO Connection DATA	3.3V
GPIO_04	PIN_B3	GPIO Connection DATA	3.3V
GPIO_05	PIN_B4	GPIO Connection DATA	3.3V
GPIO_06	PIN_A4	GPIO Connection DATA	3.3V
GPIO_07	PIN_B5	GPIO Connection DATA	3.3V
GPIO_08	PIN_A5	GPIO Connection DATA	3.3V
GPIO_09	PIN_D5	GPIO Connection DATA	3.3V
GPIO_010	PIN_B6	GPIO Connection DATA	3.3V
GPIO_011	PIN_A6	GPIO Connection DATA	3.3V
GPIO_012	PIN_B7	GPIO Connection DATA	3.3V
GPIO_013	PIN_D6	GPIO Connection DATA	3.3V
GPIO_014	PIN_A7	GPIO Connection DATA	3.3V
GPIO_015	PIN_C6	GPIO Connection DATA	3.3V
GPIO_016	PIN_C8	GPIO Connection DATA	3.3V
GPIO_017	PIN_E6	GPIO Connection DATA	3.3V
GPIO_018	PIN_E7	GPIO Connection DATA	3.3V
GPIO_019	PIN_D8	GPIO Connection DATA	3.3V
GPIO_020	PIN_E8	GPIO Connection DATA	3.3V
GPIO_021	PIN_F8	GPIO Connection DATA	3.3V
GPIO_022	PIN_F9	GPIO Connection DATA	3.3V
GPIO_023	PIN_E9	GPIO Connection DATA	3.3V
GPIO_024	PIN_C9	GPIO Connection DATA	3.3V
GPIO_025	PIN_D9	GPIO Connection DATA	3.3V
GPIO_026	PIN_E11	GPIO Connection DATA	3.3V
GPIO_027	PIN_E10	GPIO Connection DATA	3.3V
GPIO_028	PIN_C11	GPIO Connection DATA	3.3V
GPIO_029	PIN_B11	GPIO Connection DATA	3.3V
GPIO_030	PIN_A12	GPIO Connection DATA	3.3V
GPIO_031	PIN_D11	GPIO Connection DATA	3.3V
GPIO_032	PIN_D12	GPIO Connection DATA	3.3V
GPIO_033	PIN_B12	GPIO Connection DATA	3.3V

Table 3-7 GPIO-1 Pin Assignments

Signal Name	*FPGA Pin No.*	*Description*	*I/O Standard*
GPIO_1_IN0	PIN_T9	GPIO Connection DATA	3.3V
GPIO_10	PIN_F13	GPIO Connection DATA	3.3V
GPIO_1_IN1	PIN_R9	GPIO Connection DATA	3.3V
GPIO_11	PIN_T15	GPIO Connection DATA	3.3V
GPIO_12	PIN_T14	GPIO Connection DATA	3.3V
GPIO_13	PIN_T13	GPIO Connection DATA	3.3V
GPIO_14	PIN_R13	GPIO Connection DATA	3.3V
GPIO_15	PIN_T12	GPIO Connection DATA	3.3V

GPIO_16	PIN_R12	GPIO Connection DATA	3.3V
GPIO_17	PIN_T11	GPIO Connection DATA	3.3V
GPIO_18	PIN_T10	GPIO Connection DATA	3.3V
GPIO_19	PIN_R11	GPIO Connection DATA	3.3V
GPIO_110	PIN_P11	GPIO Connection DATA	3.3V
GPIO_111	PIN_R10	GPIO Connection DATA	3.3V
GPIO_112	PIN_N12	GPIO Connection DATA	3.3V
GPIO_113	PIN_P9	GPIO Connection DATA	3.3V
GPIO_114	PIN_N9	GPIO Connection DATA	3.3V
GPIO_115	PIN_N11	GPIO Connection DATA	3.3V
GPIO_116	PIN_L16	GPIO Connection DATA	3.3V
GPIO_117	PIN_K16	GPIO Connection DATA	3.3V
GPIO_118	PIN_R16	GPIO Connection DATA	3.3V
GPIO_119	PIN_L15	GPIO Connection DATA	3.3V
GPIO_120	PIN_P15	GPIO Connection DATA	3.3V
GPIO_121	PIN_P16	GPIO Connection DATA	3.3V
GPIO_122	PIN_R14	GPIO Connection DATA	3.3V
GPIO_123	PIN_N16	GPIO Connection DATA	3.3V
GPIO_124	PIN_N15	GPIO Connection DATA	3.3V
GPIO_125	PIN_P14	GPIO Connection DATA	3.3V
GPIO_126	PIN_L14	GPIO Connection DATA	3.3V
GPIO_127	PIN_N14	GPIO Connection DATA	3.3V
GPIO_128	PIN_M10	GPIO Connection DATA	3.3V
GPIO_129	PIN_L13	GPIO Connection DATA	3.3V
GPIO_130	PIN_J16	GPIO Connection DATA	3.3V
GPIO_131	PIN_K15	GPIO Connection DATA	3.3V
GPIO_132	PIN_J13	GPIO Connection DATA	3.3V
GPIO_133	PIN_J14	GPIO Connection DATA	3.3V

3.6 A/D Converter and 2x13 Header

The DE0-Nano contains an ADC128S022 lower power, eight-channel CMOS 12-bit analog-to-digital converter. This A-to-D provides conversion throughput rates of 50 ksps to 200 ksps. It can be configured to accept up to eight input signals at inputs IN0 through IN7. This eight input signals are connected to the 2x13 header, as shown in Figure 3-10. The remaining I/Os of the 2x13 header are a DC +3.3V (VCC33), a GND and 13 pins, which are connect directly to the Cyclone IV E device.

For more detailed information on the A/D converter chip, please refer to its datasheet which is available on manufacturer's website or under the /datasheet folder of the system CD.

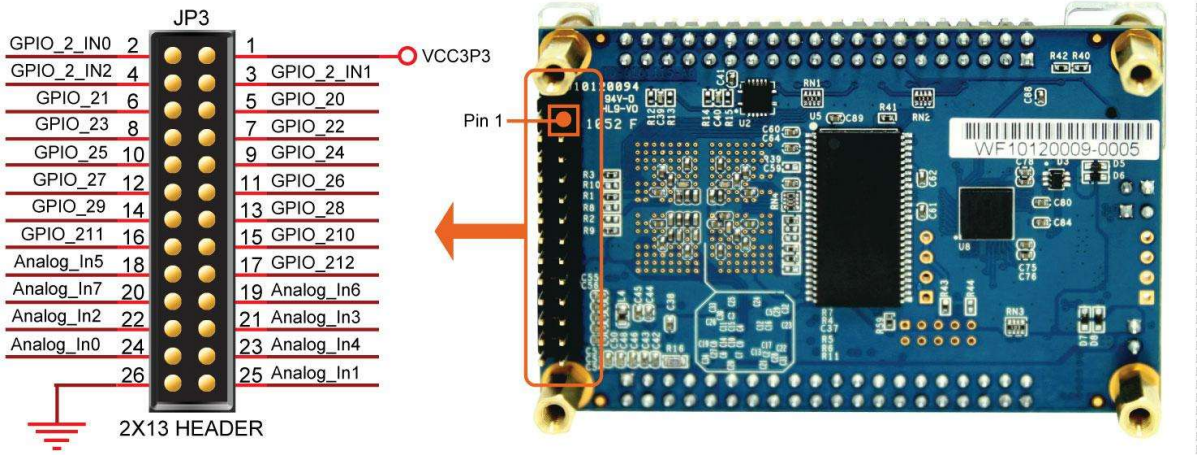


Figure 3-10 Pin distribution of the 2x13 Header

Figure 3-11 shows the connections on the 2x13 header, A/D converter and Cyclone IV device.

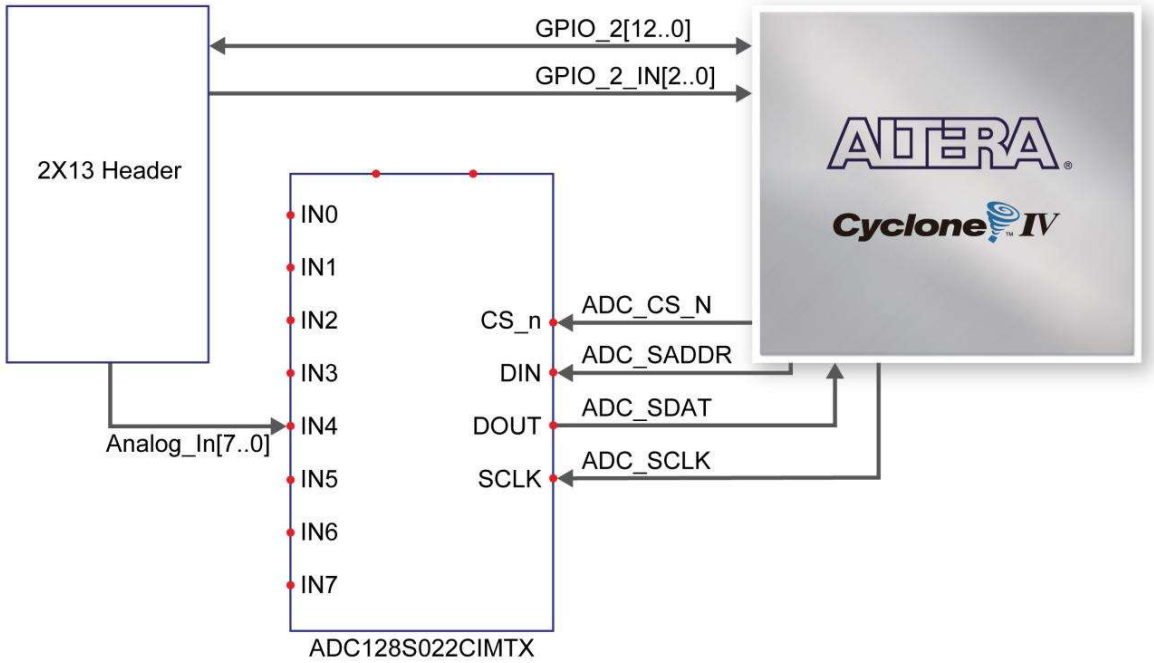


Figure 3-11 Wiring for 2x13 header and A/D converter

The pictures below indicate the pin 1 location of the 2x13 header.

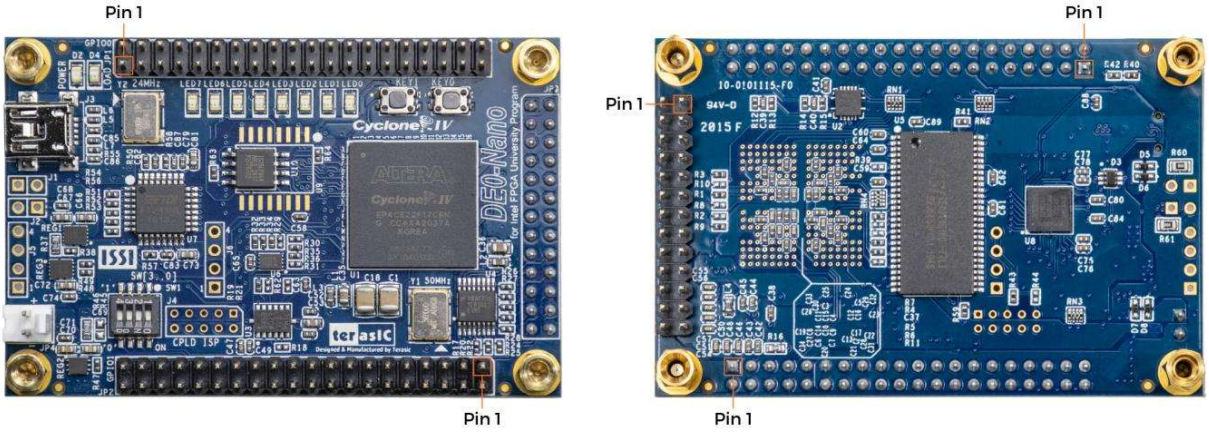


Figure 3-12 Pin1 locations of the 2x13 header

Table 3-8 Pin Assignments for 2x13 Header

Signal Name	*FPGA Pin No.*	*Description*	*I/O Standard*
GPIO_2[0]	PIN_A14	GPIO Connection DATA[0]	3.3V
GPIO_2[1]	PIN_B16	GPIO Connection DATA[1]	3.3V
GPIO_2[2]	PIN_C14	GPIO Connection DATA[2]	3.3V
GPIO_2[3]	PIN_C16	GPIO Connection DATA[3]	3.3V
GPIO_2[4]	PIN_C15	GPIO Connection DATA[4]	3.3V
GPIO_2[5]	PIN_D16	GPIO Connection DATA[5]	3.3V
GPIO_2[6]	PIN_D15	GPIO Connection DATA[6]	3.3V
GPIO_2[7]	PIN_D14	GPIO Connection DATA[7]	3.3V
GPIO_2[8]	PIN_F15	GPIO Connection DATA[8]	3.3V
GPIO_2[9]	PIN_F16	GPIO Connection DATA[9]	3.3V
GPIO_2[10]	PIN_F14	GPIO Connection DATA[10]	3.3V
GPIO_2[11]	PIN_G16	GPIO Connection DATA[11]	3.3V
GPIO_2[12]	PIN_G15	GPIO Connection DATA[12]	3.3V
GPIO_2_IN[0]	PIN_E15	GPIO Input	3.3V
GPIO_2_IN[1]	PIN_E16	GPIO Input	3.3V
GPIO_2_IN[2]	PIN_M16	GPIO Input	3.3V

Table 3-9 Pin Assignments for ADC

Signal Name	*FPGA Pin No.*	*Description*	*I/O Standard*
ADC_CS_N	PIN_A10	Chip select	3.3V
ADC_SADDR	PIN_B10	Digital data input	3.3V
ADC_SDAT	PIN_A9	Digital data output	3.3V
ADC_SCLK	PIN_B14	Digital clock input	3.3V